

STA347 — Probability

Course Notes

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Contents

Lecture 1: Set Theory Fundamentals

Sep 8, 2026

“Probability = Set Theory + Measure”

1. Basic Set Operations

Union, Intersect, Complement, ...

$A \cup B$, $A \cap B$, A^C , ...

2. Def 1.1.1 Convergence of Real-Valued Sequences

A sequence $(x_n)_{n=1}^{\infty} \subseteq \mathbb{R}$ converges to x if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } n > N \Rightarrow |x_n - x| < \varepsilon.$$

$x_n \rightarrow x$ *converge to x ?*

Pick ε (how close x_n is to the target).

For any ε picked, can find N , such that $n > N$, $|x_n - x| < \varepsilon$.

e.g. $x_n = \frac{1}{n} \rightarrow 0$

3. Def 1.1.2 Functional Convergence Pointwisely

A sequence $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$ converges pointwisely to f (*target f*) if

$$\forall \varepsilon > 0, \forall x \in X, \exists N_x \in \mathbb{N} \text{ s.t. } n > N_x \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

$f_1, \dots, f_n, \dots$ — *sequence of functions*

f — *target function*

For any ε and x , can find N_x s.t. $n > N_x$, $|f_n(x) - f(x)| < \varepsilon$.

$f_1(2), f_2(2), \dots, f_n(2), \dots \rightarrow f(2)$ — *for every x , there is a corresponding N_x .*

4. Def 1.1.3 Functional Convergence Uniformly

A sequence $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$ converges uniformly to f if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \quad \forall x \in X, n > N \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

That works for all $x \in X$ — need a common threshold N for uniform convergence.

Key difference:

For pointwise convergence, you need to have x specified first, then there is a threshold N_x corresponding to each x .

For uniform convergence, you need a common threshold N that works for all $x \in X$.

Example 1

$$f_n(x) = \frac{x}{n+x}, \quad x \geq 0$$

Then for any fixed value x , as $n \rightarrow \infty$, $f_n(x) \rightarrow 0$:

$$x = 1: f_n(1) = \frac{1}{n+1} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$x = 2: f_n(2) = \frac{2}{n+2} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\therefore f_n(x) \xrightarrow{\text{pointwise}} 0$$

However, f_n does not converge uniformly to 0: no N works for all x .

No matter how you specify N , there is always a larger value of x for which the bound fails.

$$\Rightarrow f_n(x) \not\xrightarrow{\text{uniformly}} 0$$

Limiting Processes

Def 1.2.1 The Supremum of a Set

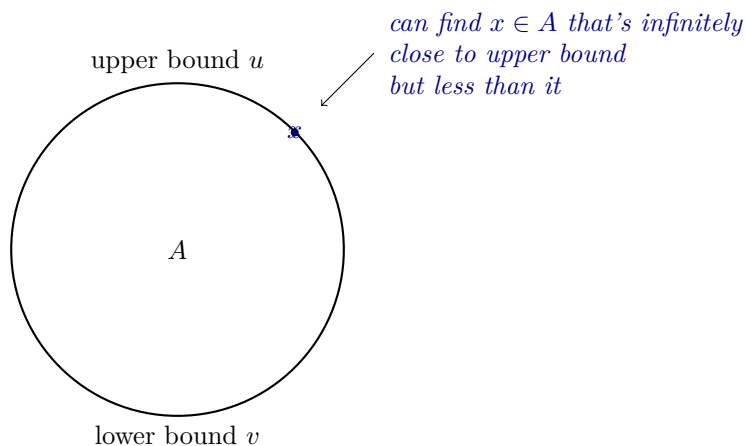
Let $A \subseteq \mathbb{R}$ be a subset. $u = \sup A$ iff

$$\forall \varepsilon > 0, \exists x \in A \quad u - x < \varepsilon \quad \text{and} \quad \forall x \in A, x \leq u.$$

Def 1.2.2 The Infimum of a Set

Let $A \subseteq \mathbb{R}$ be a subset. $v = \inf A$ iff

$$\forall \varepsilon > 0, \exists x \in A \quad x - v < \varepsilon \quad \text{and} \quad \forall x \in A, x \geq v.$$



Example 2

$$A = (0, 1)$$

$$\sup A = 1 \quad \inf A = 0$$

Distinction of sup/inf vs. max/min:

$\max(A)$ and $\min(A)$ must be elements of the set A .

$\sup(A)$ and $\inf(A)$ may not be elements of A .

Example 3

$$A = (0, 1)$$

$\max(A)$ and $\min(A)$ do not exist. $\sup(A)$ and $\inf(A)$ do exist.

$$B = (0, 1]$$

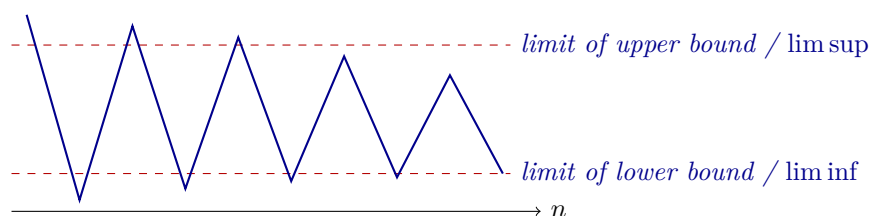
$\min(B)$ does not exist, $\max(B) = 1$. $\sup(B) = 1$, $\inf(B) = 0$.

Def 1.2.3 Limsup and Liminf

Let $(x_n) \subseteq \mathbb{R}$ be a sequence.

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \sup_{m \geq n} x_m = \inf_{n \geq 0} \sup_{m \geq n} x_m$$

$$\liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf_{m \geq n} x_m = \sup_{n \geq 0} \inf_{m \geq n} x_m$$



Example 4

$$x_n = (-1)^n = -1, 1, -1, 1, \dots$$

No limit, diverges: $\lim_{n \rightarrow \infty} x_n$ DNE.

But $\sup_{m \geq n} x_m = 1$ and $\inf_{m \geq n} x_m = -1$

$$\Rightarrow \limsup_{n \rightarrow \infty} x_n = 1, \quad \liminf_{n \rightarrow \infty} x_n = -1$$

The limsup and liminf are used to describe the long-run behaviour of a sequence, even if the sequence does not converge.

For a sequence x_n , if $\limsup_{n \rightarrow \infty} x_n = \liminf_{n \rightarrow \infty} x_n = x$, then $\lim_{n \rightarrow \infty} x_n = x$.

Function Limits

Def 1.3 Left and Right Limit of a Function

For any function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x-) = \lim_{y \uparrow x} f(y) = L \quad (\text{to the left of } x, \text{ a small neighbourhood})$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall y \in (x - \delta, x), |f(y) - L| < \varepsilon.$$

$$f(x+) = \lim_{y \downarrow x} f(y) = U$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall y \in (x, x + \delta), |f(y) - U| < \varepsilon.$$

If $f(x-) = f(x+) = C$, then $\lim_{y \rightarrow x} f(y) = C$.

Example 5

$$f(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

$$f(0-) = 0, f(0+) = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) \text{ DNE}$$

Lecture 2: Probability Measure

Sep 10, 2026

Def 2.1 Sample Space

A sample space Ω is any non-empty set.

Example 2.1 The sample space contains “all the states of the following”

Coin Flipping: $\Omega = \{H, T\}$

Weather: $\{\text{hot, cold, mild}\} \times \{\text{wet, dry}\}$

Def 2.2 Probability Measure (STA347 ver.)

For a sample space Ω , we say P is a probability measure if the following holds:

- i) $\forall E \subseteq \Omega, 0 \leq P(E) \leq 1$, that is $P : \mathcal{P}(\Omega) \rightarrow [0, 1]$
- ii) $P(\Omega) = 1$
- iii) $\forall E_1, E_2, \dots \subseteq \Omega$ such that $E_i \cap E_j = \emptyset \quad \forall i \neq j$

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

$\mathcal{P}(\Omega)$ is the power set of Ω .

Remark: It may be impossible for such a P to exist. To define exactly which subsets we want P to satisfy these axioms, that would require more advanced mathematical theories. For this class, we just assume such a P is defined for all subsets of Ω .

Example 2.2 Probabilities of Coin Flipping

$$\Omega = \{H, T\}$$

$$\mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

$$P : \mathcal{P}(\Omega) \mapsto [0, 1]$$

$$P : \emptyset \mapsto 0$$

$$\{H\} \mapsto \frac{1}{2}$$

$$\{T\} \mapsto \frac{1}{2}$$

$$\{H, T\} \mapsto 1$$

Example 2.3 The Uniform/Lebesgue Measure

Let $\Omega = [L, U]$ for some $L < U \in \mathbb{R}$. Define

$$P([a, b]) = \frac{b - a}{U - L} \quad \text{when } L \leq a \leq b \leq U$$

We assume such a measure exists (and is well defined) for this class.

Proposition 2.3 The following properties are satisfied by any probability measure P

1. $\forall E \subseteq \Omega, P(E^C) = 1 - P(E)$
2. $P(\phi) = 0$
3. $\forall E, F \subseteq \Omega$, such that $E \subseteq F, P(E) \leq P(F)$ (*“big event, big probability”*)
4. $\forall E, F \subseteq \Omega, P(E \cup F) = P(E) + P(F) - P(E \cap F)$
5. $\forall E, F \subseteq \Omega, P(E \cap F) = P(E) + P(F) - P(E \cup F)$
6. $\forall E, F \subseteq \Omega, P(F \setminus E) = P(F) - P(E \cap F)$

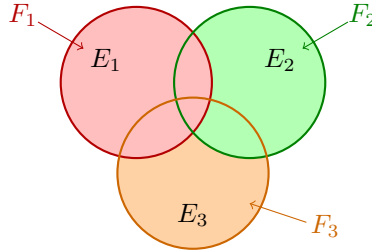
Proof via Def 2.2 (probability axioms).

Lemma 2.4 For a sequence $E_1, \dots, E_n \subseteq \Omega$

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i)$$

Proof. We define $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j, F_1 = E_1$

$$\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i \text{ and } F_i \cap F_j = \phi \text{ if } i \neq j$$



$F_i = \text{the part of } E_i \text{ not already covered by } E_1, \dots, E_{i-1}.$

from the probability axiom.

$$\begin{aligned} \text{Since } \bigcup_{i=1}^n E_i &= \bigcup_{i=1}^n F_i \\ P\left(\bigcup_{i=1}^n E_i\right) &= P\left(\bigcup_{i=1}^n F_i\right) \\ &= \sum_{i=1}^n P(F_i) \end{aligned}$$

We also know $F_i \subseteq E_i$ as $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j$

$$\begin{aligned} P(F_i) &\leq P(E_i) \\ P\left(\bigcup_{i=1}^n E_i\right) &= \sum_{i=1}^n P(F_i) \leq \sum_{i=1}^n P(E_i) \end{aligned}$$

□

Lemma 2.5 Let P be the Lebesgue measure on $[0, 1]$. Then $P(E) = 0$ for any countable set E .

Lecture 3: Independence, Random Variables and Distributions

Sep 15, 2026

Def 2.7 Independence

Events $E_1, \dots, E_n \subseteq \Omega$ are independent with respect to a probability measure P , if for each $\mathcal{I} \subseteq \begin{matrix} [n] \\ \text{any subset of } \{1, \dots, n\} \end{matrix}$

$$P\left(\bigcap_{i \in \mathcal{I}} E_i\right) = \prod_{i \in \mathcal{I}} P(E_i)$$

e.g. if $\mathcal{I} = \{1, 2, 3\}$ a subset of $\{1, \dots, n\}$

$$P(E_1 \cap E_2 \cap E_3) = P(E_1)P(E_2)P(E_3)$$

Proposition 2.7.1 If $E_1, \dots, E_n$ are independent, then $E_1^c, \dots, E_n^c$ are also independent. Proof as an exercise.

Proposition 2.7.2 If an infinite collection of events $\{E_\alpha : \alpha \in \mathcal{I}\}$ are independent, then for a finite subset $\mathcal{J} \subseteq \mathcal{I}$, the events $\{E_\beta : \beta \in \mathcal{J}\}$ are independent.

Random Variables and Distributions

Def. 3.1 Random Variables

Given a sample space Ω , a function $X : \Omega \rightarrow \mathbb{R}$ is a random variable.

Example 3.1.1 Coin Flipping

$$\Omega = \{H, T\}$$

$$\downarrow \text{R.V. } X : \quad X(H) \rightarrow 1, \quad X(T) \rightarrow 0$$

$$\mathbb{R}$$

Theorem 3.2 Distribution

A random variable X and probability measure P induce a probability measure μ on $\mathbb{R}$ defined by

$$\mu(A) = P(X \in A) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

for any $A \subseteq \mathbb{R}$. *μ is a measure on the real number set: $\mu(\{0, 1\})$ and $\mu([0, 1])$ each return a certain value.*

$\mu(A)$ takes a real number set A as input, and equals P of the *pre-image of A w.r.t. r.v. X* , i.e. $P(\Omega_A) = \mu(A)$ where $\Omega_A = \{\omega \in \Omega : X(\omega) \in A\}$.

Example 3.2.1 Define the following

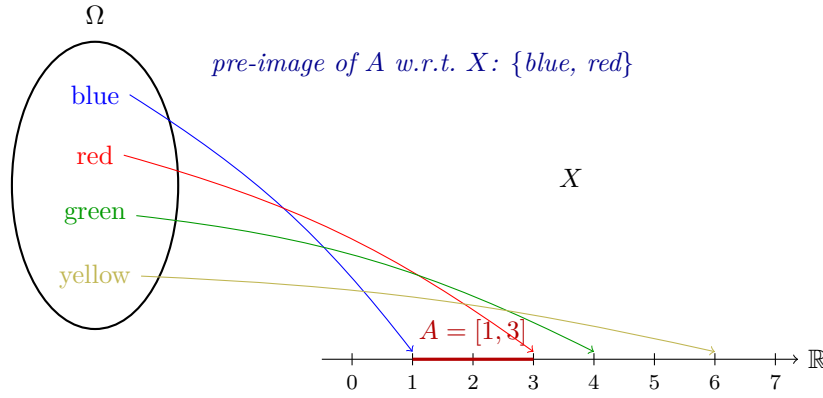
$$\Omega = \{\text{blue}, \text{red}, \text{green}, \text{yellow}\}$$

$$X : \begin{cases} \text{blue} \rightarrow 1 \\ \text{red} \rightarrow 3 \\ \text{green} \rightarrow 4 \\ \text{yellow} \rightarrow 6 \end{cases}$$

$$P : \{\text{blue}\} \mapsto \frac{1}{5}, \quad \{\text{red}\} \mapsto \frac{1}{5}, \quad \{\text{green}\} \mapsto \frac{1}{4}, \quad \{\text{yellow}\} \mapsto 1 - \frac{1}{5} - \frac{1}{5} - \frac{1}{4} = \frac{7}{20}$$

$$P(\{\text{blue}, \text{red}\}) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

$$A = [1, 3]$$



1. $\{w : w \in \Omega, X(w) \in A\} = \{\text{blue}, \text{red}\}$
2. $P(\{\text{blue}, \text{red}\}) = \frac{2}{5}$, so $\mu(A) = \frac{2}{5}$

in summary

$$\mu([1, 3]) = \frac{2}{5}$$

Claim: μ is a probability measure on $\mathbb{R}$ (for any real number set B).

Proof. 1. $\mu(B) \in [0, 1]$ is trivial, since P is a probability measure.

2. Since $X(\omega) \in \mathbb{R}$ for all $\omega \in \Omega$.

$$\mu(\mathbb{R}) = P(\{\omega : \omega \in \Omega, X(\omega) \in \mathbb{R}\}) = P(\Omega) = 1$$

3. Consider disjoint sets $A_1, A_2, \dots \subseteq \mathbb{R}$ and define $E_i = \{\omega : X(\omega) \in A_i\}$
(e.g. $E_1 = \{\omega : X(\omega) \in A_1\}$, $E_2 = \{\omega : X(\omega) \in A_2\}$)

Since X is a function, it is impossible for $X(\omega) = a$ and $X(\omega) = b$ when $a \neq b$, so the E_i 's

are also disjoint. Then

$$\begin{aligned}
\mu\left(\bigcup_{i=1}^{\infty} A_i\right) &= P\left(X(\omega) \in \bigcup_{i=1}^{\infty} A_i\right) \\
&= P\left(\bigcup_{i=1}^{\infty} E_i\right) \quad \text{we know } E_i\text{'s are disjoint} \\
&= \sum_{i=1}^{\infty} P(E_i) \\
&= \sum_{i=1}^{\infty} \mu(A_i)
\end{aligned}$$

□

Def. 3.3 The Cumulative Distribution Function

A random variable X and measure P generate the cumulative distribution function (CDF) defined by

$$F(x) = \mu((-\infty, x]) = P(X \leq x)$$

where $P(X \leq x)$ is a short-hand notation for

$$P(\{\omega : \omega \in \Omega, X(\omega) \in (-\infty, x]\})$$

Theorem 3.3.1 A distribution function F for a random variable X uniquely defines the measure μ .

Proof is outside the scope of this class.

Example 3.3.2 Uniform Distribution

P is uniform on $[0, 1]$ and $X(\omega) = \omega$

For $x \in [0, 1)$

$$\begin{aligned}
F(x) &= \mu((-\infty, x]) && \leftarrow \text{by definition of CDF} \\
&= P(X \leq x) && \leftarrow \text{by Thm 3.2 (definition of distribution)} \\
&= P(\{\omega : X(\omega) \in (-\infty, x]\}) \\
&= P((-\infty, x]) && \leftarrow \text{by } X(\omega) = \omega \\
&= P([0, x]) && \leftarrow \text{because } P \text{ is a uniform/Lebesgue measure} \\
&= x
\end{aligned}$$

Theorem 3.4 Properties of Distribution Function

A distribution function satisfies the following properties:

- i) For all $x \leq y \in \mathbb{R}$, $F(x) \leq F(y)$
- ii) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$
- iii) F is right-continuous

Proof. i) $x \leq y \in \mathbb{R}$

$$F(x) = \mu((-\infty, x]), F(y) = \mu((-\infty, y])$$

Since $x \leq y$, $(-\infty, x] \subseteq (-\infty, y]$

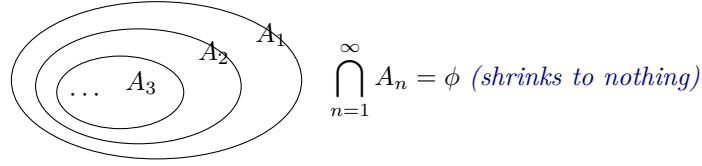
$$\mu((-\infty, x]) \leq \mu((-\infty, y])$$

“bigger set, bigger probability”

$$(A \subseteq B, \quad P(A) \leq P(B) \quad / \quad \mu(A) \leq \mu(B))$$

ii) Define $A_n = \{X \leq -n\}$, $n \geq 0$

Then $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$

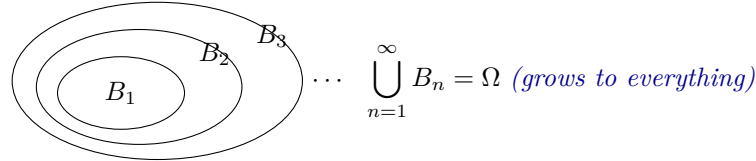


$$\bigcap_{n=1}^{\infty} A_n = \phi$$

$$\begin{aligned} \lim_{n \rightarrow \infty} F(-n) &= \lim_{n \rightarrow \infty} P(A_n) \\ &= P\left(\bigcap_{n=1}^{\infty} A_n\right) \\ &= P(\phi) = 0 \end{aligned}$$

Define $B_n = \{X \leq n\}$,

then $B_1 \subseteq B_2 \subseteq \dots$



$$\bigcup_{n=1}^{\infty} B_n = \Omega$$

$$\begin{aligned} \lim_{n \rightarrow \infty} F(n) &= \lim_{n \rightarrow \infty} P(B_n) \\ &= P\left(\bigcup_{n=1}^{\infty} B_n\right) \\ &= P(\Omega) = 1 \end{aligned}$$

iii) Fix $x \in \mathbb{R}$ and let $h_n \downarrow 0$ ($h_n \geq 0$, $h_n \rightarrow 0$, $h_1 \geq h_2 \geq \dots$)

Define $C_n = \{X \leq x + h_n\}$

then $C_1 \supseteq C_2 \supseteq \dots$

$$\lim_{n \rightarrow \infty} C_n = \bigcap_{n=1}^{\infty} C_n = \{X \leq x\}$$

Therefore

$$\lim_{n \rightarrow \infty} F(x + h_n) = \lim_{n \rightarrow \infty} P(C_n) = P\left(\bigcap_{n=1}^{\infty} C_n\right) = P(X \leq x) = F(x)$$

Hence

$$\lim_{h \downarrow 0} F(x + h) = F(x)$$

So F is right-continuous.

□

Def. 3.5 The Inverse CDF

The inverse CDF of a r.v. X is defined by

$$F^{-1}(y) = \sup \{x : F(x) < y\}$$

Theorem 3.5.1 If F satisfies properties i) to iii) of Thm 3.4, then such an F is the CDF of some r.v.

Theorem 3.6 If F is a CDF, it can only have countably many discontinuities.

Lecture 4: Random Variables & Distributions

Sep 17, 2026

Def. 3.7 Density Function

A random variable has a density function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad \text{if for all } x \in \mathbb{R}$$

$$F(x) = \int_{-\infty}^x f(y) dy$$

we call $f(\cdot)$ the density function.

Def. 3.8 Joint Distribution

We define the joint distribution function of a random vector $\vec{X} = (X_1, \dots, X_n)$ by

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

using the notation $P(A, B) \stackrel{\text{def}}{=} P(A \cap B)$

Theorem 3.8.1 A joint distribution function $F : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the following properties:

i) if $(x_1, \dots, x_n)$ and $(x'_1, \dots, x'_n)$ satisfy $x_i \leq x'_i$ for all $i \in [n]$,

$$F(x_1, \dots, x_n) \leq F(x'_1, \dots, x'_n)$$

ii) $\lim_{x_i \rightarrow -\infty} F(x_1, \dots, x_n) = 0$ for all $i \in \{1, \dots, n\}$ and $\lim_{x_1 \rightarrow \infty, \dots, x_n \rightarrow \infty} F(x_1, \dots, x_n) = 1$

iii) $\lim_{h \rightarrow 0^+} F(x_1 + h, \dots, x_n) = \dots = \lim_{h \rightarrow 0^+} F(x_1, \dots, x_n + h)$

$$= F(x_1, \dots, x_n)$$

Proofs are analogous to the proofs under the univariate case.

Def. 3.9 The Joint Density

A random vector has a joint density function:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \text{if } \forall (x_1, \dots, x_n) \in \mathbb{R}^n$$

$$F(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f(y_1, \dots, y_n) dy_n \dots dy_1$$

Theorem 3.10 Independence of Random Variables

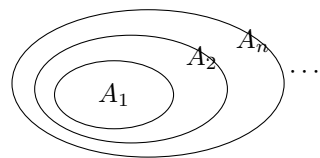
Random variables $X_1, \dots, X_n$ with CDFs $F_1, \dots, F_n$ are independent if and only if

$$F(x_1, \dots, x_n) = \prod_{i=1}^n F_i(x_i), \quad \forall (x_1, \dots, x_n) \in \mathbb{R}^n$$

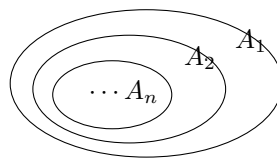
Limits of sets:

Let A_i be sets for $i \in \{1, 2, \dots\}$ and consider $\lim_{n \rightarrow \infty} A_n$.

- If $A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n \subseteq \cdots$ (increasing), then $\lim_{n \rightarrow \infty} \bigcup_{i=1}^n A_i = \bigcup_{i=1}^{\infty} A_i$.
- If $A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots$ (decreasing), then $\lim_{n \rightarrow \infty} \bigcap_{i=1}^n A_i = \bigcap_{i=1}^{\infty} A_i$.



increasing: $A_1 \subseteq A_2 \subseteq \cdots$



decreasing: $A_1 \supseteq A_2 \supseteq \cdots$

Lecture 5: Expectation

Sep 22, 2026

Expectation

In STA237:

$$E(X) = \begin{cases} \sum_x x \cdot p(x) & \text{if } X \text{ discrete} \\ \int x f(x) dx & \text{if } X \text{ continuous} \end{cases}$$

for this course, we are defining probability via “measure theory”, so we need to revisit the definition of expectation.

Particularly, under the “measure theory” perspective:

$$E(X) = \int x dP \Rightarrow P = \text{probability measure}$$

$$\int x f(x) dx \Rightarrow \int x dP$$

“Riemann Integral” (*integrating against a measure dP instead of dx ; this is the Lebesgue integral*)

Def. 5.1 Simple Expectation

If X only takes finitely many values $x_1, \dots, x_n$, and $A_i = \{\omega : X(\omega) = x_i\}$, define the expectation of X by:

$$E(X) = \sum_{i=1}^n x_i P(A_i)$$

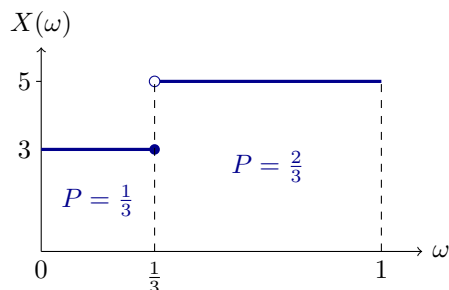
Example 5.1.1 For an arbitrary set $A \subset \Omega$, let

$$Y(\omega) = I\{\omega \in A\} \quad \text{Then } E(Y) = P(A)$$

$$\begin{aligned} E(Y) &= \sum_{i=0}^1 y_i \cdot P(A_i) \quad Y \text{ only takes two values 0 and 1} \\ &= 1 \cdot P\{\omega \in A\} + 0 \cdot P\{\omega \notin A\} \\ &= P(A) = P(\{\omega : \omega \in A\}) \end{aligned}$$

Example 5.1.2 P is uniform measure on $\Omega = [0, 1)$, and

$$X(\omega) = \begin{cases} 5 & \omega > \frac{1}{3} \\ 3 & \omega \leq \frac{1}{3} \end{cases}$$



$$\begin{aligned}
E(X) &= 5 \cdot P(\omega > \frac{1}{3}) + 3 \cdot P(\omega \leq \frac{1}{3}) \\
&= 5 \cdot (1 - \frac{1}{3}) + 3 \cdot \frac{1}{3} \\
&= \frac{13}{3}
\end{aligned}$$

Proposition 5.2 Properties of Expectation

If X and Y are simple random variables, the following hold:

- i) If $X \geq 0$ a.s., $E(X) \geq 0$
- ii) For any $a \in \mathbb{R}$, $E(aX) = a \cdot E(X)$
- iii) $E(X + Y) = E(X) + E(Y)$

Remark: "a.s." means almost surely. if E a.s. $\Rightarrow P(E) = 1$

$X \geq 0$ a.s. $\Rightarrow P(X \geq 0) = 1 \Leftrightarrow P(\{\omega : X(\omega) \geq 0\}) = 1$

we may have certain ω 's s.t. $X(\omega) < 0$, but $P(\{\omega : X(\omega) < 0\}) = 0$

Proof. Both i) and ii) are trivial.

For iii), let X take values $x_1, \dots, x_n$ on sets $A_1, \dots, A_n$ (i.e. $A_i = \{\omega : X(\omega) = x_i\}$) and Y takes values $y_1, \dots, y_m$ on sets $B_1, \dots, B_m$.

Observe that there are $m \cdot n$ events

$$C_{ij} = \{X = x_i \cap Y = y_j\}$$

Then

$$\begin{aligned}
E(X + Y) &= \sum_{i=1}^n \sum_{j=1}^m (x_i + y_j) \cdot P(C_{ij}) \\
&= \sum_{i=1}^n x_i \sum_{j=1}^m P(X = x_i \cap Y = y_j) + \sum_{j=1}^m y_j \sum_{i=1}^n P(X = x_i \cap Y = y_j) \\
&= \sum_{i=1}^n x_i P(X = x_i) + \sum_{j=1}^m y_j P(Y = y_j) \\
&= E(X) + E(Y)
\end{aligned}$$

□

Lemma 5.2.1 If Proposition 5.2 holds, the following properties also hold:

- iv) If $X \leq Y$ a.s., then $E(X) \leq E(Y)$
- v) If $X = Y$ a.s., then $E(X) = E(Y)$
- vi) $|E(X)| \leq E(|X|)$

proof as exercise.

Our logic is, we first define

expectation of simple random variables
 $\Downarrow$
 expectation of bounded R.V.'s (approximation)
 $\Downarrow$
 expectation of non-negative R.V.'s (truncation)
 $\Downarrow$
 expectation of any R.V.'s (symmetric)

Def. 5.3 Bounded Random Variable

A random variable X is bounded if there exists $M < \infty$, such that $|X| \leq M$ a.s.

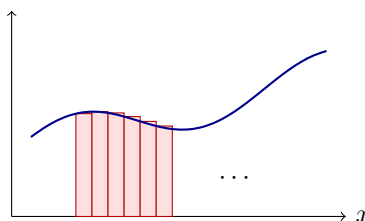
Def. 5.3.1 Expectation of Bounded Random Variable

If X is a bounded R.V., then

$$\begin{aligned}
 E(X) &= \inf \{E(Z) : Z \text{ simple, } Z \geq X \text{ a.s.}\} \\
 &= \sup \{E(Y) : Y \text{ simple, } Y \leq X \text{ a.s.}\}
 \end{aligned}$$

Recall, when we defined the integral:

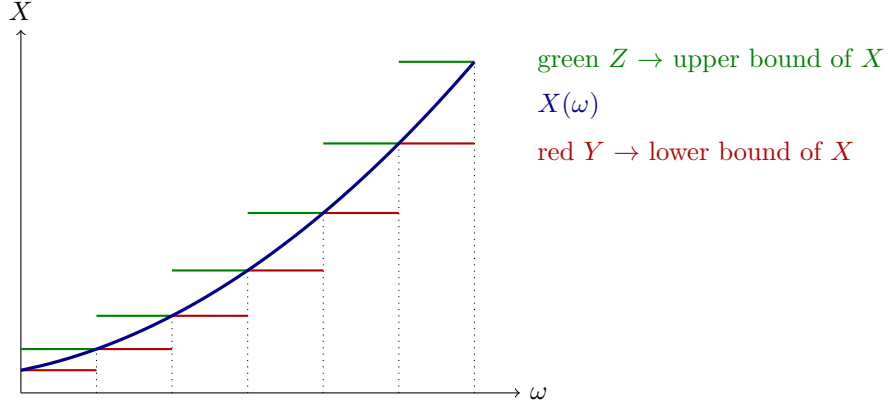
the integral is defined as the limit value of sums of areas of tiny rectangles.



When it comes to the measure-theoretic integral, we are no longer cutting X axis into tiny intervals, but cutting each set into tiny subsets and calculate the sum of x times the measure of each tiny subset.

(Prof's notes say "Riemann" here, but cutting the sample space into sets and weighting by their measure is the Lebesgue integral; Riemann is the "cut the x-axis" picture above.)

$$\begin{aligned}
 E(X) &= \inf \{E(Z) : Z \text{ is simple and } Z \geq X \text{ a.s.}\} \\
 &= \sup \{E(Y) : Y \text{ is simple and } Y \leq X \text{ a.s.}\}
 \end{aligned}$$



“Squeeze the R.V. X via two simple R.V.’s Y and Z , with one capturing the upper bound and the other capturing the lower bound”.

Proof. Since X is bounded, $|X| < M$ a.s.

$$L = \sup \{EY : Y \text{ simple and } Y \leq X \text{ a.s.}\}$$

$$U = \inf \{EZ : Z \text{ simple and } Z \geq X \text{ a.s.}\}$$

$$L \leq U \text{ as } Y \leq X \text{ a.s. and } Z \geq X \text{ a.s.}$$

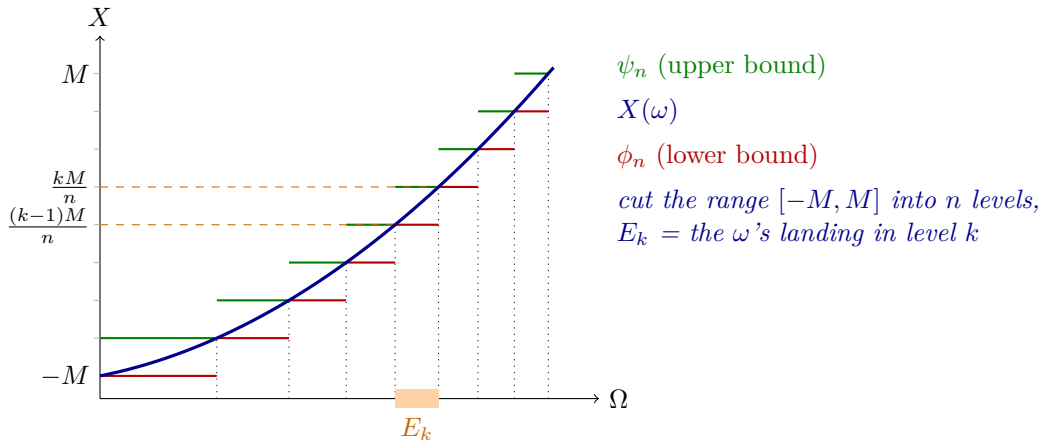
Our goal is to prove $L \geq U \Rightarrow L = U \stackrel{\text{def}}{=} E(X)$

the way to cut $[-M, M]$ into tiny intervals.

for any arbitrary n , we define

$$E_k = \left\{ \omega : \frac{kM}{n} \geq X(\omega) \geq \frac{(k-1)M}{n} \right\}$$

($k = -n+1, \dots, n$, so the E_k ’s cover $[-M, M]$ in $2n$ levels of width $\frac{M}{n}$.)



we then define two simple R.V.’s

Lower bound:

$$\phi_n(\omega) = \sum_k \frac{(k-1)M}{n} \cdot I(\omega \in E_k)$$

Upper bound:

$$\psi_n(\omega) = \sum_k \frac{kM}{n} \cdot I(\omega \in E_k)$$

$$\phi_n(\omega) \leq X(\omega) \leq \psi_n(\omega)$$

$$E\phi_n(\omega) \leq E(X) \leq E\psi_n(\omega)$$

$$E\psi_n(\omega) = E\phi_n(\omega) + \frac{M}{n}$$

Let $n \rightarrow \infty$, $\frac{M}{n} \rightarrow 0$, so the upper bound and lower bound converge:

$$L \geq E\phi_n = E\psi_n - \frac{M}{n} \geq U - \frac{M}{n} \Rightarrow L \geq U$$

$$E(X) = E\psi_n(\omega) = E\phi_n(\omega) \quad \text{as } n \rightarrow \infty$$

□

$X \leq Y$ a.s. means:

step 1. $\omega : X(\omega) \leq Y(\omega)$

step 2. $\{\omega : X(\omega) \leq Y(\omega)\}$

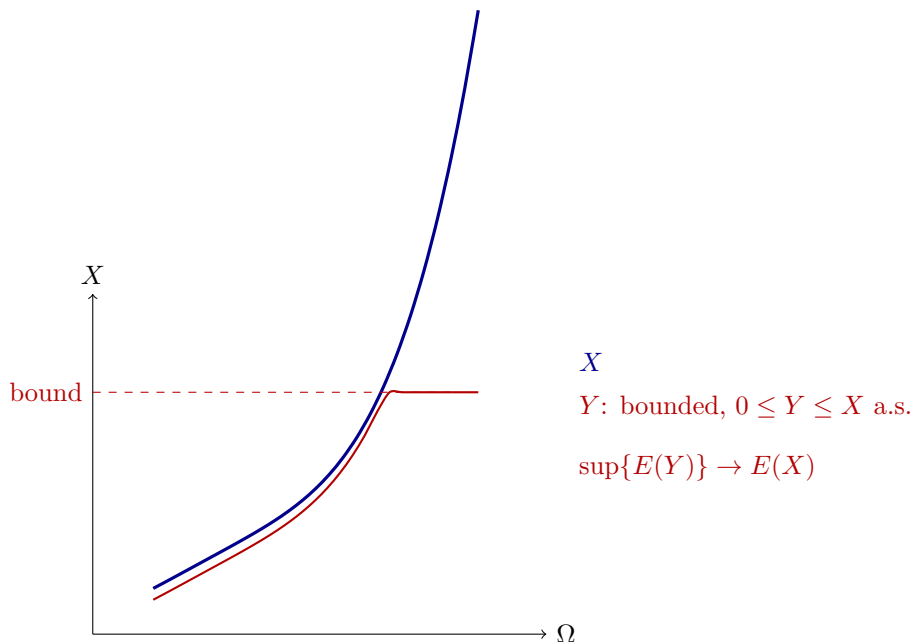
step 3. $P(\{\omega : X(\omega) \leq Y(\omega)\}) = 1$

then we say $X \leq Y$ a.s.

Def. 5.4 Non-negative Random Variable

If $X \geq 0$ a.s. ($P(\{\omega : X(\omega) \geq 0\}) = 1$)

define $E(X) = \sup \{E(Y) : Y \text{ bounded, } 0 \leq Y \leq X \text{ a.s.}\}$



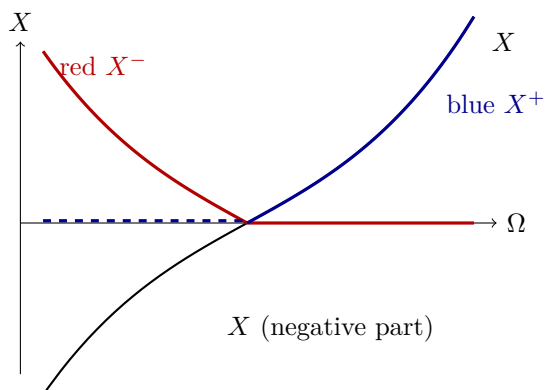
Remark: Def 5.4. only provides a definition of $E(X)$ when X is non-negative, it does not guarantee $E(X) < \infty$.

Def. 5.5 Integrable Random Variables

For any random variable X , let $X^+(\omega) = \max\{X(\omega), 0\}$, and $X^-(\omega) = \max\{-X(\omega), 0\}$.

Observe that both X^+ and X^- are non-negative and $X = X^+ - X^-$.

Then, if $E(|X|) = E(X^+) + E(X^-) < \infty$, we say R.V. X is integrable and define $E(X) = E(X^+) - E(X^-)$



1. $E|X| = E(X^+) + E(X^-) < \infty \Rightarrow X$ is integrable
2. $E(X) = E(X^+) - E(X^-)$

Proposition 5.6 Properties of Expected Values

If R.V. X is either bounded, non-negative or integrable, Proposition 5.2 holds.

1. X and Y may not need to be simple, if they are bounded, non-negative or integrable, the 3 properties (prop 5.2) also hold.
2. Lemma 5.2.1 says, if prop 5.2 holds, the additional 3 properties also hold.
3. To sum up, if R.V.'s are bounded, non-negative or integrable, all the 6 properties hold.

Hint for Assignment 1. Q2:

E_1 and E_2 are independent.

E_1^c and E_2^c are independent.

if $E_1, E_2, \dots, E_n$, you want to prove $E_1^c, E_2^c, \dots, E_n^c$ are independent.

$$P(E_1^c \cap E_2^c \cap \dots \cap E_n^c) = \dots$$

you can flip 1 event at a time

first try to flip the first...

$$\begin{aligned} & P(E_1^c \cap E_2 \cap E_3 \cap \dots \cap E_n) \\ &= P(E_2 \cap E_3 \cap \dots \cap E_n) - P(E_1 \cap E_2 \cap \dots \cap E_n) \end{aligned}$$

$$= \cdots = (1 - P(E_1))P(E_2) \cdots P(E_n)$$

Then you flip the second one...

$$P(E_1^c \cap E_2^c \cap E_3 \cap \cdots \cap E_n)$$

Via induction, complete the proofs

Lecture 6: Independence and Expectation, Variance

Sep 24, 2026

Theorem 5.7 Independence and Expectation

If X and Y are independent with $EX, EY < \infty$, then

$$E(XY) = (EX)(EY)$$

Proof. First, consider both X and Y are simple, so that

$$X = \sum_{i=1}^n x_i I(A_i) \quad \text{and} \quad Y = \sum_{j=1}^m y_j I(B_j)$$

$$\begin{aligned} E(XY) &= E\left[\sum_{i=1}^n \sum_{j=1}^m x_i y_j \cdot I\{A_i \cap B_j\}\right] \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j \underbrace{P(X \in A_i, Y \in B_j)}_{= P(X \in A_i) \cdot P(Y \in B_j) \text{ because } X \text{ and } Y \text{ indep.}} \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j P(\{\omega : X(\omega) \in A_i, Y(\omega) \in B_j\}) \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j P(X \in A_i) \cdot P(Y \in B_j) \\ &= \underbrace{\sum_{i=1}^n x_i \cdot P(X \in A_i)}_{EX} \underbrace{\sum_{j=1}^m y_j \cdot P(Y \in B_j)}_{EY} \\ &= (EX)(EY) \end{aligned}$$

Next, consider the case that X and Y are bounded and non-negative, then we let $Z_1 \geq X, Z_2 \geq Y$, where Z_1 and Z_2 are simple. Notice that

$$Z_1 \geq X, Z_2 \geq Y \text{ implies } Z_1 \cdot Z_2 \geq XY$$

$$\begin{aligned} (EX)(EY) &= \inf \{EZ_1 : Z_1 \text{ simple, } Z_1 \geq X \text{ a.s.}\} \\ &\quad \cdot \inf \{EZ_2 : Z_2 \text{ simple, } Z_2 \geq Y \text{ a.s.}\} \\ &= \inf \{(EZ_1) \cdot (EZ_2)\} \quad \text{because } Z_1, Z_2 \text{ simple, so } \inf(EZ_1) \cdot \inf(EZ_2) = \inf(EZ_1 \cdot EZ_2) \\ &= \inf \{EZ_1 Z_2 : Z_1 \text{ and } Z_2 \text{ simple and independent, } Z_1 Z_2 \geq XY \text{ a.s.}\} \end{aligned}$$

Additionally, if Z_1 and Z_2 are simple and

$$Z_1 \leq X \text{ a.s.} \quad Z_2 \leq Y \text{ a.s.}$$

Similarly,

$$(EX)(EY) = \sup \{EZ_1Z_2 : Z_1 \text{ and } Z_2 \text{ are simple and independent, } Z_1 \leq X, Z_2 \leq Y \text{ a.s.}\}$$

Previously (Def 5.3.1), we proved that

$$\begin{aligned} & \inf \{EZ_1Z_2 : Z_1, Z_2 \text{ are simple and independent with } Z_1 \geq X, Z_2 \geq Y \text{ a.s.}\} \\ &= \sup \{EZ_1Z_2 : Z_1, Z_2 \text{ are simple and independent with } Z_1 \leq X, Z_2 \leq Y \text{ a.s.}\} = E(XY) \\ &\Rightarrow (EX)(EY) = E(XY) \end{aligned}$$

Finally, for X and Y that are integrable, first realize that if X and Y are independent, then X^+, X^-, Y^+, Y^- are also independent.

$$X = X^+ - X^- \quad Y = Y^+ - Y^-$$

$$\begin{aligned} E(XY) &= E[(X^+ - X^-)(Y^+ - Y^-)] \\ &= E[X^+Y^+ - X^+Y^- - X^-Y^+ + X^-Y^-] \\ &= E(X^+Y^+) - E(X^+Y^-) - E(X^-Y^+) + E(X^-Y^-) \\ &= E(X^+)E(Y^+) - E(X^+)E(Y^-) - E(X^-)E(Y^+) + E(X^-)E(Y^-) \\ &= (EX)(EY) \end{aligned}$$

□

Def. 5.8 Variance and Covariance

The variance of a random variable X is defined as

$$\text{Var}(X) = E[(X - EX)^2]$$

The covariance of random variables X and Y is

$$\text{Cov}(X, Y) = E[(X - EX)(Y - EY)]$$

The correlation coefficient is

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Lecture 7: Chapter 4 - Inequalities

Sep 29, 2026

Theorem 4.1 Markov's Inequality

If $X \geq 0$ a.s., then for all $a > 0$,

$$P(X \geq a) \leq \frac{E(X)}{a}$$

Proof. Define $Z(\omega) = a \cdot I\{X(\omega) \geq a\}$. Then, $Z \leq X$ a.s., and since Z is simple,

$$E(Z) = a \cdot P(X \geq a)$$

$$P(X \geq a) = \frac{E(Z)}{a} \leq \frac{E(X)}{a}$$

□

□

Theorem 4.2 Chebyshev's Inequality

For all $t > 0$,

$$P(|X - E(X)| \geq t) \leq \frac{\text{Var}(X)}{t^2}$$

Proof. Apply Markov's inequality to $Y = [X - E(X)]^2 \geq 0$:

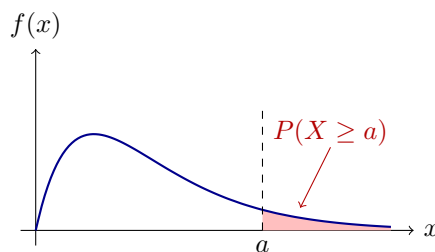
$$P(|X - E(X)| \geq t) = P([X - E(X)]^2 \geq t^2) \leq \frac{E[(X - E(X))^2]}{t^2} = \frac{\text{Var}(X)}{t^2}$$

□

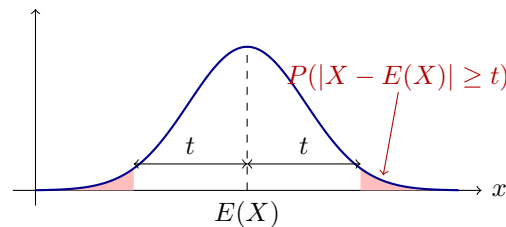
□

Both Chebyshev and Markov's inequality talk about

$$P(X \geq \dots) \Leftarrow \text{the tail probability}$$



Markov: uses the 1st moment



Chebyshev: uses the 2nd moment

"Concentration of Probability Mass"

Where Markov uses the 1st moment, Chebyshev uses 2nd moment.

What if we use other transformation/moments to quantify the tail probability?

E.g. the exponential moment? $\Rightarrow$ Chernoff's Inequality

$$M_X(t) = E[e^{tX}]$$

Theorem 4.3 Chernoff's Inequality

For all $t \geq 0$, $P(X \geq E(X) + t) \leq \inf_{\lambda > 0} M_{X-E(X)}(\lambda) e^{-\lambda t}$

$$\begin{aligned} P(X \geq E(X) + t) &\text{ roughly } P(\underbrace{X - E(X)}_{\substack{\text{distance between } X \\ \text{to its expected value}}} \geq t) \\ &\leq \inf_{\lambda > 0} M_{X-E(X)}(\lambda) e^{-\lambda t} \end{aligned}$$

Proof. the function $g(x) = e^x$ is monotonically increasing and non-negative.

$$\begin{aligned} &P(X - E(X) \geq t) \\ &= P(e^{X-E(X)} \geq e^t) \\ &= P(e^{\lambda(X-E(X))} \geq e^{\lambda t}) \quad \lambda > 0 \end{aligned}$$

Since $e^{\lambda(X-E(X))} \geq 0$ a.s., we can apply Markov's inequality

$$\leq \frac{E[e^{\lambda(X-E(X))}]}{e^{\lambda t}} \quad \text{where the numerator is the mgf } M_{X-E(X)}(\lambda)$$

We proved that

$$P(X - E(X) \geq t) \leq e^{-\lambda t} \cdot M_{X-E(X)}(\lambda)$$

Since this inequality holds for any arbitrary $\lambda > 0$, we can take the inf:

$$P(X - E(X) \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t} M_{X-E(X)}(\lambda)$$

□

□

Hoeffding's Inequality (Thm 4.4)

Def. 4.4.1 Convex Function

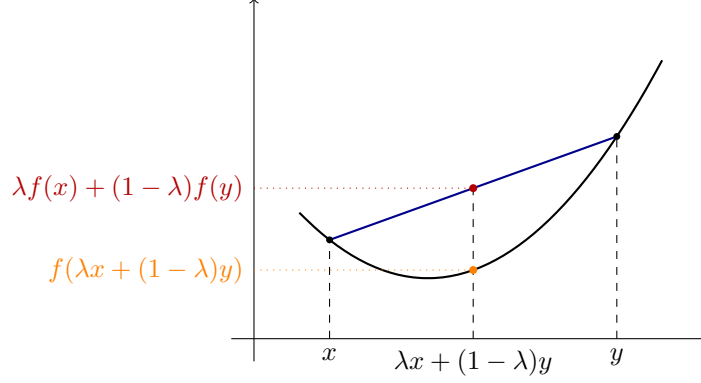
A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex if for all $\lambda \in (0, 1)$ and $x, y \in \mathbb{R}$

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Any value between x and y can be represented as

$$\begin{aligned} &\lambda x + (1 - \lambda)y \text{ for } \lambda \in (0, 1) \\ &\overbrace{\lambda x + (1 - \lambda)y, \lambda \in (0, 1)} \end{aligned}$$

if $\lambda \rightarrow 1$, then the value is closer to x . Vice versa for $\lambda \rightarrow 0$, the value is closer to y .



Convexity: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$
 $\Rightarrow$ *the function is under the straight line*

Lemma 4.4.2 Hoeffding's Lemma

If Y satisfies $E(Y) = 0$ and $a \leq Y \leq b$ a.s., then for all $\lambda > 0$

$$M_Y(\lambda) \leq e^{\lambda^2(b-a)^2/8}$$

Proof. Since $a \leq Y \leq b$ a.s., $Y = \alpha \cdot a + (1 - \alpha) \cdot b$.

Transforming $Y \Rightarrow \alpha$, where $0 \leq \alpha \leq 1$,

$$\alpha = \frac{b - Y}{b - a}, \quad 1 - \alpha = \frac{Y - a}{b - a}$$

Also, since the function $e^{\lambda y}$ is convex,

$$e^{\lambda(\alpha a + (1-\alpha)b)} \leq \alpha e^{\lambda a} + (1 - \alpha)e^{\lambda b}$$

Apply $f(y) = e^{\lambda y}$ transformation to r.v. Y :

$$e^{\lambda Y} \leq \alpha e^{\lambda a} + (1 - \alpha)e^{\lambda b} = \frac{b - Y}{b - a} e^{\lambda a} + \frac{Y - a}{b - a} e^{\lambda b}$$

Take expectation on both sides:

$$\begin{aligned} E[e^{\lambda Y}] &\leq E\left[\frac{b - \overbrace{Y}^{\text{random var.}}}{b - a} \underbrace{e^{\lambda a}}_{\text{constant}}\right] + E\left[\frac{Y - a}{b - a} \underbrace{e^{\lambda b}}_{\text{constant}}\right] \\ &= \frac{b - E(Y)}{b - a} e^{\lambda a} + \frac{E(Y) - a}{b - a} e^{\lambda b} \\ &= \underbrace{\frac{b}{b - a}}_p e^{\lambda a} + \frac{-a}{b - a} e^{\lambda b} \quad (\text{since } E(Y) = 0) \end{aligned}$$

Next, we let $p = \frac{b}{b-a}$ (*since $E(Y) = 0$ forces $a \leq 0 \leq b$, we have $p \in [0, 1]$*):

$$E[e^{\lambda Y}] \leq p e^{\lambda a} + (1 - p) e^{\lambda b}$$

We take log on RHS:

$$\begin{aligned}
\log\{E[e^{\lambda Y}]\} &\leq \log\{pe^{\lambda a} + (1-p)e^{\lambda b}\} \\
&= \log\{e^{\lambda a}[p + (1-p)e^{\lambda(b-a)}]\} \\
&= \underbrace{\lambda a}_{=(p-1)u} + \log[p + (1-p)e^{\lambda(b-a)}] \\
(p-1)u &= \left(\frac{b}{b-a} - 1\right)(b-a)\lambda = \frac{a}{b-a}(b-a)\lambda = a\lambda
\end{aligned}$$

Next, let $u = (b-a)\lambda$:

$$\begin{aligned}
\log\{E[e^{\lambda Y}]\} &\leq (p-1)u + \log[p + (1-p)e^u] =: \varphi(u) \\
\varphi(0) &= 0, \quad \varphi'(u) = (p-1) + \frac{(1-p)e^u}{p + (1-p)e^u}, \quad \varphi'(0) = 0 \\
\varphi(u) &= \underbrace{\varphi(0)}_0 + \underbrace{\varphi'(0)}_0 u + \frac{1}{2}\varphi''(\xi)u^2 \quad (\text{Taylor's Expansion at } u=0)
\end{aligned}$$

For some ξ between 0 and u , $\varphi(u) = \frac{1}{2}\varphi''(\xi)u^2$

$$\varphi''(u) = \frac{p(1-p)e^u}{[p + (1-p)e^u]^2} \leq \frac{1}{4}$$

(writing $q = \frac{(1-p)e^u}{p+(1-p)e^u} \in [0, 1]$, $\varphi''(u) = q(1-q) \leq \frac{1}{4}$)

$$\begin{aligned}
&\Rightarrow \varphi(u) \leq \frac{1}{8}u^2 \\
&\Rightarrow \log\{E[e^{\lambda Y}]\} \leq \frac{1}{8}u^2 = \frac{(b-a)^2\lambda^2}{8} \quad \text{as } u = (b-a)\lambda \\
&\Rightarrow M_Y(\lambda) \leq e^{\frac{(b-a)^2\lambda^2}{8}}
\end{aligned}$$

□

□

Theorem 4.4 Hoeffding's Inequality

Suppose $X_1, X_2, \dots, X_n, \dots$ are indep. with $a_i \leq X_i \leq b_i$ for all i .

Then letting $S_n = \sum_{i=1}^n X_i$, for all $t > 0$,

$$P(|S_n - E(S_n)| \geq t) \leq 2 \cdot \exp\left\{-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right\}$$

Proof. We first prove the $P(S_n - E(S_n) \geq t)$ part.

Applying Chernoff's inequality,

$$P(\underbrace{S_n - E(S_n)}_{(1)} \geq t) \leq e^{-\lambda t} \underbrace{E[e^{\lambda(S_n - E(S_n))}]}_{(2)}$$

$$(1): \quad S_n - E(S_n) = \sum_{i=1}^n (X_i - E(X_i)) \quad \text{as } S_n = \sum_{i=1}^n X_i$$

(2):

$$e^{\lambda(S_n - E(S_n))} = e^{\lambda \sum_{i=1}^n (X_i - E(X_i))} = \prod_{i=1}^n e^{\lambda[X_i - E(X_i)]}$$

Taking expectations, the factors are independent so E of the product = product of E 's (*Thm 5.7*):

$$M_{S_n - E(S_n)}(\lambda) = \prod_{i=1}^n M_{X_i - E(X_i)}(\lambda)$$

Applying Hoeffding's Lemma to $Y = X_i - E(X_i)$ (*mean 0, range of width $b_i - a_i$*): $M_{X_i - E(X_i)}(\lambda) \leq e^{\frac{(b_i - a_i)^2 \lambda^2}{8}}$

$$M_{S_n - E(S_n)}(\lambda) \leq \prod_{i=1}^n e^{\frac{(b_i - a_i)^2 \lambda^2}{8}} = e^{\frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2}$$

Combine with Chernoff's inequality:

$$P(S_n - E(S_n) \geq t) \leq e^{-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2}$$

Rearrange, we have Hoeffding's inequality.

Filling in the rearrangement: write $D = \sum_{i=1}^n (b_i - a_i)^2$. The bound holds for every $\lambda > 0$, so pick the λ minimizing the exponent $g(\lambda) = -\lambda t + \frac{\lambda^2}{8} D$:

$$g'(\lambda) = -t + \frac{\lambda}{4} D = 0 \Rightarrow \lambda^* = \frac{4t}{D}, \quad g(\lambda^*) = -\frac{4t^2}{D} + \frac{2t^2}{D} = -\frac{2t^2}{D}$$

$$\Rightarrow P(S_n - E(S_n) \geq t) \leq e^{-2t^2/D}$$

The other tail: apply the same argument to $-X_i$ (with $-b_i \leq -X_i \leq -a_i$, same width $b_i - a_i$) to get $P(S_n - E(S_n) \leq -t) \leq e^{-2t^2/D}$. Then by Lemma 2.4 (union bound),

$$P(|S_n - E(S_n)| \geq t) \leq P(S_n - E(S_n) \geq t) + P(S_n - E(S_n) \leq -t) \leq 2e^{-2t^2/D}$$

□

□

Lecture 8:

Oct 1, 2026

Theorem 4.5 Jensen's Inequality

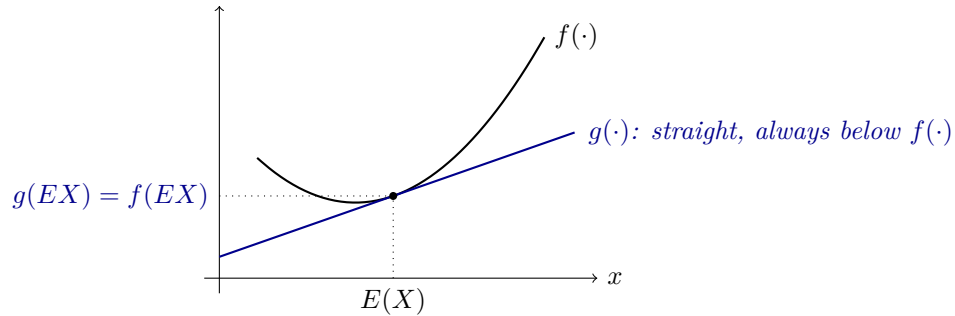
If f is a convex function and X is a random variable such that $f(X)$ is integrable,

$$f(EX) \leq Ef(X)$$

The inequality is **flipped** if f is **concave**.

Proof. Since f is convex, there exists a function g s.t.

$$\begin{aligned} g(x) &= ax + b \text{ and} && \Leftrightarrow g(\cdot) \text{ is a straight line} \\ g(x) &\leq f(x) \text{ for all } x \text{ and} && \Leftrightarrow g(\cdot) \text{ always below } f(\cdot) \\ g(EX) &= f(EX) && \Leftrightarrow g(\cdot) = f(\cdot) \text{ at } x = EX \end{aligned}$$



$$f(x) \geq g(x) \text{ for all } x$$

$$\begin{aligned} E[f(X)] &\geq E[g(X)] = E[aX + b] \\ &= aE(X) + b \\ &= g(E(X)) \\ &= f(E(X)) \end{aligned}$$

□

Example 4.5.1 Applying the Jensen's Inequality

$$\begin{aligned} [E(X)]^2 &\leq E[X^2] \quad f(x) = x^2 \Rightarrow \text{convex} \\ E[\log X] &\leq \log[EX] \quad f(x) = \log x \Rightarrow \text{concave} \end{aligned}$$

Lemma 4.6 Young's Inequality

For any $p, q > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$,

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad \text{for any } x, y \geq 0$$

Proof. Let $h_y(x) = \frac{1}{p}x^p + \frac{1}{q}y^q - xy$
To prove $h_y(x) \geq 0$, we take derivatives

$$h'_y(x) = \frac{dh_y(x)}{dx} = x^{p-1} - y$$

$$h''_y(x) = \frac{d^2h_y(x)}{dx^2} = (p-1)x^{p-2} \geq 0, \text{ because } x \geq 0$$

Therefore $h'_y(x) = 0 \Rightarrow x^* = y^{\frac{1}{p-1}}$ is the minimizer.

Plugging in,

$$\begin{aligned} h_y(x^*) &= \frac{1}{p} \left(y^{\frac{1}{p-1}} \right)^p + \frac{1}{q} y^q - y^{\frac{1}{p-1}} \cdot y \\ &= \frac{1}{p} y^{\frac{p}{p-1}} + \frac{1}{q} y^q - y^{\frac{p}{p-1}} \end{aligned}$$

Since

$$\frac{1}{p} + \frac{1}{q} = 1 \Rightarrow \frac{1}{q} = 1 - \frac{1}{p} \Rightarrow q = \frac{p}{p-1}$$

substituting $q = \frac{p}{p-1}$ and $\frac{1}{q} = 1 - \frac{1}{p}$:

$$\begin{aligned} h_y(x^*) &= \frac{1}{p} y^{\frac{p}{p-1}} + \left(1 - \frac{1}{p} \right) y^{\frac{p}{p-1}} - y^{\frac{p}{p-1}} \\ &= 0 \end{aligned}$$

$$\Rightarrow h_y(x) \geq h_y(x^*) = 0$$

Therefore $\frac{1}{p}x^p + \frac{1}{q}y^q \geq xy$.

(If $p = q = 2$: $\frac{1}{2}x^2 + \frac{1}{2}y^2 \geq xy \Leftrightarrow x^2 + y^2 \geq 2xy$.) □

Theorem 4.6 Hölder's Inequality

If $p, q > 1$ s.t. $\frac{1}{p} + \frac{1}{q} = 1$,

$$E[|XY|] \leq [E(|X|^p)]^{\frac{1}{p}} \cdot [E(|Y|^q)]^{\frac{1}{q}}$$

Proof. Let

$$X^* = \frac{|X|}{[E(|X|^p)]^{\frac{1}{p}}} \quad \text{and} \quad Y^* = \frac{|Y|}{[E(|Y|^q)]^{\frac{1}{q}}}$$

Observe that

$$E[(X^*)^p] = E[(Y^*)^q] = 1$$

since

$$E[(X^*)^p] = E \left[\left(\frac{|X|}{[E(|X|^p)]^{\frac{1}{p}}} \right)^p \right] = E \left[\frac{|X|^p}{E(|X|^p)} \right] = \frac{E(|X|^p)}{E(|X|^p)} = 1$$

and the same for Y^* with q .

Applying the Young's Inequality

$$X^* Y^* \leq \frac{(X^*)^p}{p} + \frac{(Y^*)^q}{q}$$

Take expectations on both sides

$$E[X^*Y^*] \leq \frac{E[(X^*)^p]}{p} + \frac{E[(Y^*)^q]}{q}$$

$$E \left[\frac{|X||Y|}{E(|X|^p)^{\frac{1}{p}} E(|Y|^q)^{\frac{1}{q}}} \right] \leq \frac{1}{p} + \frac{1}{q} = 1$$

Therefore, re-arrange

$$E[|X||Y|] \leq E(|X|^p)^{\frac{1}{p}} E(|Y|^q)^{\frac{1}{q}}$$

□

Proposition 4.6.1 Cauchy-Schwarz Inequality

$$E[|XY|] \leq \sqrt{E(X^2)E(Y^2)}$$

Proof. Exercise. (Take $p = q = 2$ in Hölder's inequality: $E[|XY|] \leq [E(X^2)]^{\frac{1}{2}} [E(Y^2)]^{\frac{1}{2}}$.)

□

Theorem 4.7 Minkowski's Inequality

If $p > 1$, then

$$[E(|X + Y|^p)]^{\frac{1}{p}} \leq [E(|X|^p)]^{\frac{1}{p}} + [E(|Y|^p)]^{\frac{1}{p}}$$

(The triangle inequality in probability)

No proof was given in lecture.

EXAM UPDATE: CAN BRING CHEATSHEET DOUBLE SIDE HANDWRITTEN!!!!

Lecture 9: Chapter 5 - Limiting Theorems and Convergences

Oct 6, 2026

Theorem 5.1 Continuity of Measure

If $\{A_n\} \uparrow A$ or $\{A_n\} \downarrow A$, then $\lim_{n \rightarrow \infty} P(A_n) = P(A)$.

Note:

1. $\{A_n\} \uparrow A$ means:

$$A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n \subseteq \cdots \rightarrow A, \quad \lim_{n \rightarrow \infty} \bigcup_{i=1}^n A_i = A$$

2. $\{A_n\} \downarrow A$ means:

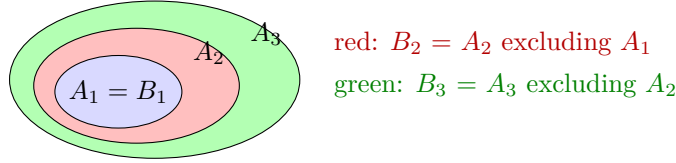
$$A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots \rightarrow A, \quad \lim_{n \rightarrow \infty} \bigcap_{i=1}^n A_i = A$$

Note: $A_1 \subseteq \cdots \subseteq A_n \subseteq \cdots \rightarrow A$ means

$$P(A_1) \leq \cdots \leq P(A_n) \leq \cdots \rightarrow P(A)$$

Proof. Suppose $\{A_n\} \uparrow A$.

We let $A_0 = \emptyset$, and $B_1 = A_1$, $B_n = A_n \cap A_{n-1}^c$ ($B_n =$ the part of A_n not already in A_{n-1}).



Observe that:

1. $\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n B_i$
2. B_i 's are disjoint.

Therefore, $P(\bigcup_{i=1}^n A_i) = P(\bigcup_{i=1}^n B_i)$, and

$$\begin{aligned} P(A) &= P\left(\lim_{n \rightarrow \infty} \bigcup_{i=1}^n A_i\right) = P\left(\lim_{n \rightarrow \infty} \bigcup_{i=1}^n B_i\right) = P\left(\bigcup_{i=1}^{\infty} B_i\right) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n P(B_i) && \text{(countable additivity, } B_i \text{'s disjoint)} \\ &= \lim_{n \rightarrow \infty} P(A_n) && (\sum_{i=1}^n P(B_i) = P(\bigcup_{i=1}^n B_i) = P(A_n)) \end{aligned}$$

For the case $A_n \downarrow A$, the proof is similar; one may take the complement:

$$\{A_n\} \downarrow A \Rightarrow \{A_n^c\} \uparrow A^c$$

$$\text{so } P(A_n) = 1 - P(A_n^c) \rightarrow 1 - P(A^c) = P(A).$$

□

Example 5.1.1 If each $A_n = [0, 1 - \frac{1}{n}]$, then

$$\lim_{n \rightarrow \infty} A_n = [0, 1) = A, \quad A_n \uparrow A$$

If we apply a uniform measure $P(\cdot)$ on $\{A_n\}$,

$$P(A) = \lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1$$

Def. 5.2 Tail Events

Consider a sequence $A_1, A_2, \dots \subseteq \Omega$, define the tail events as:

$$\limsup_{n \rightarrow \infty} A_n = \{A_n \text{ i.o.}\} = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$$

$$\liminf_{n \rightarrow \infty} A_n = \{A_n \text{ a.a.}\} = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$$

Note: the notations

i.o. means **infinitely often**

a.a. means **almost always**

$$\limsup_{n \rightarrow \infty} A_n = \{A_n \text{ i.o.}\} = \bigcap_{n=1}^{\infty} \underbrace{\bigcup_{k=n}^{\infty} A_k}_{\text{tail union}}$$

You want to find the smallest set that contains the union of your tail events:

“elements that will appear infinitely many times”.

$\omega \in \{A_n \text{ i.o.}\} \Leftrightarrow \text{for every } n \text{ there is some } k \geq n \text{ with } \omega \in A_k.$

$$\liminf_{n \rightarrow \infty} A_n = \{A_n \text{ a.a.}\} = \bigcup_{n=1}^{\infty} \underbrace{\bigcap_{k=n}^{\infty} A_k}_{\text{tail intersect}}$$

You want to find the largest set that contains the intersect of your tail events:

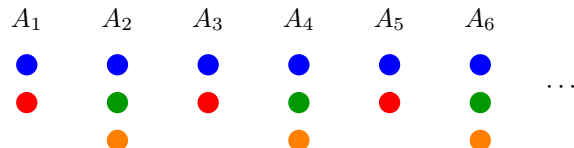
“elements that will appear in all tail sets”.

$\omega \in \{A_n \text{ a.a.}\} \Leftrightarrow \text{there is some } n \text{ with } \omega \in A_k \text{ for all } k \geq n \text{ (in all but finitely many } A_k).$

Example Tail Events

Suppose $\{A_n\}$ is constructed as follows:

$$A_i = \begin{cases} \{\text{blue, red}\} & i \text{ odd} \\ \{\text{blue, green, orange}\} & i \text{ even} \end{cases}$$



$$\limsup_{n \rightarrow \infty} A_n = \{A_n \text{ i.o.}\} = \{\text{blue, red, green, orange}\}$$

$$\liminf_{n \rightarrow \infty} A_n = \{A_n \text{ a.a.}\} = \{\text{blue}\}$$

$\{A_n \text{ i.o.}\} \Rightarrow$ “the elements that will appear infinitely often in the tail of $\{A_n\}$ ”

$\{A_n \text{ a.a.}\} \Rightarrow$ “the elements that will appear almost always in the tail of $\{A_n\}$ ”

Corollary 5.2.1

$$P(A_n \text{ i.o.}) = 1 - P(A_n^c \text{ a.a.})$$

(By De Morgan, $\{A_n \text{ i.o.}\}^c = \bigcup_n \bigcap_{k \geq n} A_k^c = \{A_n^c \text{ a.a.}\}.$)

Proposition 5.2.2

$$P(A_n \text{ a.a.}) \leq \liminf_{n \rightarrow \infty} P(A_n) \leq \limsup_{n \rightarrow \infty} P(A_n) \leq P(A_n \text{ i.o.})$$

i.e.

$$P\left(\liminf_{n \rightarrow \infty} A_n\right) \leq \liminf_{n \rightarrow \infty} P(A_n) \leq \limsup_{n \rightarrow \infty} P(A_n) \leq P\left(\limsup_{n \rightarrow \infty} A_n\right)$$

Proof. We prove the 1st “ $\leq$ ”.

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \underbrace{\bigcap_{k=n}^{\infty} A_k}_{B_n}$$

We let $B_n = \bigcap_{k=n}^{\infty} A_k$ (“the intersect of tail events”), so $\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} B_n$.

Notice that $B_1 \subseteq B_2 \subseteq \dots \subseteq B_n \subseteq \dots$ as $B_n = \bigcap_{k=n}^{\infty} A_k$:

$$\begin{aligned} B_1 &= A_1 \cap A_2 \cap \dots \cap A_n \cap \dots \\ B_2 &= A_2 \cap \dots \cap A_n \cap \dots \quad \Rightarrow B_1 \subseteq B_2 \end{aligned}$$

So $B_n \uparrow B = \bigcup_{n=1}^{\infty} B_n$, and $\lim_{n \rightarrow \infty} P(B_n) = P(B)$ by continuity of measure (Thm 5.1).

$$\begin{aligned} P\left(\liminf_{n \rightarrow \infty} A_n\right) &= P\left(\bigcup_{n=1}^{\infty} B_n\right) = P\left(\lim_{n \rightarrow \infty} B_n\right) \\ &= \lim_{n \rightarrow \infty} P(B_n) && \text{as } B_n \text{ converges} \\ &= \liminf_{n \rightarrow \infty} P(B_n) = \liminf_{n \rightarrow \infty} P\left(\bigcap_{k=n}^{\infty} A_k\right) \\ &\leq \liminf_{n \rightarrow \infty} P(A_n) && \text{because } \bigcap_{k=n}^{\infty} A_k \subseteq A_n \end{aligned}$$

The 2nd “ $\leq$ ” is by definition.

The 3rd “ $\leq$ ” can be proved similarly.

(Let $C_n = \bigcup_{k=n}^{\infty} A_k$, so $C_n \downarrow \limsup A_n$. Then $P(\limsup A_n) = \lim P(C_n) = \limsup P(C_n) \geq \limsup P(A_n)$, since $A_n \subseteq C_n$.) $\square$

Theorem 5.3 Borel-Cantelli Lemma i) If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then $P(A_n \text{ i.o.}) = 0$.

ii) If $\sum_{n=1}^{\infty} P(A_n) = \infty$ and $\{A_n\}$ are independent, then $P(A_n \text{ i.o.}) = 1$.

Proof. i)

$$\{A_n \text{ i.o.}\} = \bigcap_{n=1}^{\infty} \underbrace{\bigcup_{k=n}^{\infty} A_k}_{B_n}, \quad B_n = \bigcup_{k=n}^{\infty} A_k, \quad B_n \downarrow B$$

$$\begin{aligned} P(A_n \text{ i.o.}) &= P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = P\left(\bigcap_{n=1}^{\infty} B_n\right) \\ &= \lim_{n \rightarrow \infty} P(B_n) && \text{as } B_n \downarrow B = \bigcap_{n=1}^{\infty} B_n \\ &= \lim_{n \rightarrow \infty} P\left(\bigcup_{k=n}^{\infty} A_k\right) \\ &\leq \lim_{n \rightarrow \infty} \sum_{k=n}^{\infty} P(A_k) && (\text{union bound, Lemma 2.4}) \\ &= 0 && \text{as } \sum_{k=1}^{\infty} P(A_k) < \infty \end{aligned}$$

(The tail $\sum_{k=n}^{\infty} P(A_k)$ of a convergent series goes to 0.)

ii)

$$\begin{aligned} 1 - P(A_n \text{ i.o.}) &= P(\{A_n \text{ i.o.}\}^c) \\ &= P\left(\bigcup_{n=1}^{\infty} \underbrace{\bigcap_{k=n}^{\infty} A_k^c}_{C_n}\right) && \text{because } \{A_n \text{ i.o.}\}^c = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c \end{aligned}$$

Let $C_n = \bigcap_{k=n}^{\infty} A_k^c$, $C_n \uparrow C$.

$$\begin{aligned} &= P\left(\bigcup_{n=1}^{\infty} C_n\right) = P\left(\lim_{n \rightarrow \infty} C_n\right) && \text{as } C_n \uparrow C \\ &= \lim_{n \rightarrow \infty} P(C_n) = \lim_{n \rightarrow \infty} P\left(\bigcap_{k=n}^{\infty} A_k^c\right) \\ &= \lim_{n \rightarrow \infty} \prod_{k=n}^{\infty} [1 - P(A_k)] && \text{because } \{A_n\} \text{ independent} \end{aligned}$$

($P(\bigcap_{k=n}^{\infty} A_k^c) = \prod_{k=n}^{\infty} P(A_k^c)$; the A_k^c are independent by Prop 2.7.1.)

Each factor $1 - P(A_k) \leq 1$, but that alone doesn't give 0. The missing step: use $1 - x \leq e^{-x}$:

$$\prod_{k=n}^{\infty} [1 - P(A_k)] \leq \prod_{k=n}^{\infty} e^{-P(A_k)} = \exp\left\{-\sum_{k=n}^{\infty} P(A_k)\right\} = e^{-\infty} = 0$$

since $\sum_{k=n}^{\infty} P(A_k) = \infty$ for every n . So

$$1 - P(A_n \text{ i.o.}) = 0 \Rightarrow P(A_n \text{ i.o.}) = 1$$

□

Example 5.3.1 Converse does NOT hold for i)

Uniform measure and $A_n = \left[0, \frac{1}{n}\right]$. Then $A_n \downarrow A = \{0\}$, and

$$\begin{aligned}\{A_n \text{ i.o.}\} &= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \left[0, \frac{1}{k}\right] \\ &= \{0\} = \lim_{n \rightarrow \infty} A_n\end{aligned}$$

$$P(\{A_n \text{ i.o.}\}) = P(\{0\}) = 0$$

but

$$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

($P(A_n \text{ i.o.}) = 0$ does not force $\sum P(A_n) < \infty$.)

Example 5.3.2 Independence is needed for ii)

Define $C_1, C_2, \dots$ to be results of a fair coin toss, such that $C_i = 1$ if the i -th toss resulted in H.

Let $A_1, A_2, \dots$ be events such that

$$A_j = \left\{ \text{the } \text{1st} \text{ toss is H} \right\} \quad \text{for every } j$$

Then

$$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{2} = \lim_{n \rightarrow \infty} \frac{1}{2} \cdot n = \infty$$

$$P(A_n \text{ i.o.}) = P(A_n) = \frac{1}{2} \neq 1$$

as all $\{A_n\}$ are the same (*so they are not independent*).

Lecture 10: Convergence in Probability and Almost Sure Convergence

Oct 8, 2026

Def. 5.4 Convergence in Probability

A sequence of random variables X_n converges in probability to X if for all $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} P(|X_n - X| \leq \varepsilon) = 1 \quad \text{or} \quad \lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0$$

and is denoted by $X_n \xrightarrow{P} X$.

Def. 5.5 Almost Sure Convergence

A sequence of random variables X_n converges almost surely to X if

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

and is denoted by $X_n \xrightarrow{a.s.} X$.

Compare the two:

$$\begin{aligned} X_n \xrightarrow{P} X & \quad \lim_{n \rightarrow \infty} P(|X_n - X| < \varepsilon) = 1 \\ X_n \xrightarrow{a.s.} X & \quad P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1 \end{aligned}$$

where

$$\begin{aligned} \lim_{n \rightarrow \infty} X_n = X & \Leftrightarrow \left\{ \omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \right\} \\ P\left(\lim_{n \rightarrow \infty} X_n = X\right) &= P\left(\left\{ \omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \right\}\right) = 1 \end{aligned}$$

Written in the same form, for all $\varepsilon > 0$:

$$\begin{aligned} X_n \xrightarrow{P} X & \quad \lim_{n \rightarrow \infty} P(|X_n - X| < \varepsilon) = 1 \\ X_n \xrightarrow{a.s.} X & \quad P(|X_n(\omega) - X(\omega)| < \varepsilon \text{ a.a.}) = 1 \end{aligned}$$

“ $\xrightarrow{a.s.}$ ”: the condition holds almost always, i.e. on the intersect of the tail (for each ω , eventually every $X_n(\omega)$ stays within ε).

“ $\xrightarrow{P}$ ”: the condition only needs to be satisfied with probability tending to 1 at each large n , one n at a time.

Proposition 5.5.1 If for all $\varepsilon > 0$, $P(|X_n - X| > \varepsilon \text{ i.o.}) = 0$, then

$$X_n \xrightarrow{a.s.} X$$

Proof.

$$\begin{aligned}
P(\lim_{n \rightarrow \infty} X_n = X) &= P(\forall \varepsilon > 0, |X_n(\omega) - X(\omega)| < \varepsilon \text{ a.a.}) \\
&= 1 - P(|X_n(\omega) - X(\omega)| \geq \varepsilon \text{ i.o.}) \quad (\text{Corollary 5.2.1}) \\
&= 1 - 0 = 1
\end{aligned}$$

(Strictly: “ $\forall \varepsilon > 0$ ” is an uncountable intersection, so take $\varepsilon = \frac{1}{m}$, $m = 1, 2, \dots$; then $P(\exists m : |X_n - X| \geq \frac{1}{m} \text{ i.o.}) \leq \sum_m P(|X_n - X| \geq \frac{1}{m} \text{ i.o.}) = 0$. Also $> \varepsilon$ vs. $\geq \varepsilon$ doesn't matter since ε is arbitrary.) $\square$

Proposition 5.5.2 If for all $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} P(|X_n(\omega) - X(\omega)| \geq \varepsilon) < \infty$$

then $X_n \xrightarrow{a.s.} X$.

Proof. Applying the Borel-Cantelli Lemma.

(Let $A_n = \{|X_n - X| \geq \varepsilon\}$. By Borel-Cantelli i), $\sum P(A_n) < \infty \Rightarrow P(A_n \text{ i.o.}) = 0$ for every ε , so Prop 5.5.1 gives $X_n \xrightarrow{a.s.} X$.) $\square$

Proposition 5.5.3 If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{P} X$.

The opposite doesn't hold: $X_n \xrightarrow{P} X \nRightarrow X_n \xrightarrow{a.s.} X$.

Proof. For any $\varepsilon > 0$, let

$$E_n = \{\omega : \exists m \geq n, |X_m(\omega) - X(\omega)| \geq \varepsilon\}$$

Observe that $E_{n+1} \subseteq E_n$, and if $\omega \in \bigcap_{n=1}^{\infty} E_n$ then $X_n(\omega) \not\xrightarrow{a.s.} X(\omega)$ (it misses by $\geq \varepsilon$ infinitely often).

$$\begin{aligned}
\lim_{n \rightarrow \infty} P(|X_n(\omega) - X(\omega)| \geq \varepsilon) &\leq \lim_{n \rightarrow \infty} P(E_n) && (\{|X_n - X| \geq \varepsilon\} \subseteq E_n, \text{ take } m = n) \\
&= P\left(\bigcap_{n=1}^{\infty} E_n\right) && (\text{Thm 5.1, } E_n \downarrow) \\
&\leq P(\{\omega : X_n(\omega) \not\xrightarrow{a.s.} X(\omega)\}) && (\text{prof writes “=”; it is only } \subseteq) \\
&= 0 && \text{since } X_n \xrightarrow{a.s.} X
\end{aligned}$$

So $\lim_{n \rightarrow \infty} P(|X_n - X| \geq \varepsilon) = 0$ for every $\varepsilon > 0$, i.e. $X_n \xrightarrow{P} X$. The counterexample for the converse is Example 5.5.4. $\square$

Example 5.5.4 X_n independent with $P(X_n = 1) = \frac{1}{n}$, $P(X_n = 0) = 1 - \frac{1}{n}$.

For all $\varepsilon > 0$,

$$P(X_n > \varepsilon) = \frac{1}{n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

(for $\varepsilon \geq 1$ it is 0; either way $\rightarrow 0$)

So, $X_n \xrightarrow{P} 0$.

However,

$$P(X_n = 1 \text{ i.o.}) = 1$$

(Borel-Cantelli ii): the events $\{X_n = 1\}$ are independent and $\sum \frac{1}{n} = \infty$)

$$\Rightarrow X_n \not\xrightarrow{a.s.} 0$$

Theorem 5.5.5 If $X_n \xrightarrow{P} X$, there exists a subsequence such that

$$X_{n_k} \xrightarrow{a.s.} X$$

No proof given in lecture. Idea: pick $n_1 < n_2 < \dots$ with $P(|X_{n_k} - X| \geq \frac{1}{k}) \leq 2^{-k}$. Then $\sum_k P(|X_{n_k} - X| \geq \frac{1}{k}) < \infty$, so by Borel-Cantelli i), a.s. only finitely many $|X_{n_k} - X| \geq \frac{1}{k}$, hence $X_{n_k} \rightarrow X$ a.s.

Theorem 5.6 Continuous Mapping Theorem

If f is a continuous function, then

- i) $X_n \xrightarrow{a.s.} X \Rightarrow f(X_n) \xrightarrow{a.s.} f(X)$
- ii) $X_n \xrightarrow{P} X \Rightarrow f(X_n) \xrightarrow{P} f(X)$

Proof. i) f is continuous, it means

$$X_n(\omega) \rightarrow X(\omega) \Rightarrow f(X_n(\omega)) \rightarrow f(X(\omega))$$

so $\{\omega : X_n(\omega) \rightarrow X(\omega)\} \subseteq \{\omega : f(X_n(\omega)) \rightarrow f(X(\omega))\}$, and the left side has probability 1.

ii) For all $\varepsilon > 0$, there exists a $\delta > 0$, such that $|X_n(\omega) - X(\omega)| \leq \delta$ implies

$$|f(X_n(\omega)) - f(X(\omega))| \leq \varepsilon$$

$$\Rightarrow 1 = \lim_{n \rightarrow \infty} P(|X_n(\omega) - X(\omega)| \leq \delta) \leq \lim_{n \rightarrow \infty} P(|f(X_n(\omega)) - f(X(\omega))| \leq \varepsilon) \leq 1$$

(Prof writes “=” between the two limits; it is “ $\leq$ ”, from the event inclusion, which is all we need.)

Caveat: a single δ working for every ω needs f uniformly continuous. For general continuous f , first pick M with $P(|X| > M) < \eta$; f is uniformly continuous on $[-M-1, M+1]$, so the argument above works off an event of probability $< \eta$, and then let $\eta \rightarrow 0$.

□